

Mechanical property and micro-mechanism of Nano-SiO₂ enhanced simulated lunar soil-based geopolymer

Zhenyu Wang^a, Liqing Li^a, Faping Li^{b,*}, Banglu Xi^c

^a School of Civil Engineering and Architecture, Henan University of Technology, Zhengzhou, China

^b School of Civil Engineering, Wuhan University, Wuhan, China

^c College of Civil Engineering, Hefei University of Technology, Hefei, China

ARTICLE INFO

Keywords:

Simulated lunar soil-based geopolymer
Nano-silica
Mechanical properties
Water recovery rate
Microstructure

ABSTRACT

The lunar research station serves as critical infrastructure for establishing deep space exploration bases and achieving long-term extraterrestrial presence. To improve the mechanical properties and microstructure of simulated lunar soil-based geopolymer (SLSG), this study utilized TJ-1 lunar soil simulant as the precursor and systematically investigated the influence of Nano-SiO₂ (NS) on the water recovery rate, flowability, mechanical properties, and mass-strength efficiency of SLSG by optimizing the mixing sequence of NS and alkali activator, adjusting the water-binder ratio (0.24, 0.27, 0.30), and varying the NS dosage (0.50 wt%, 0.75 wt%, 1.00 wt%, 1.25 wt%). The enhancement mechanisms of NS were thoroughly elucidated using FT-IR, XRD, SEM, TG-DSC-DTG, EDS, and MIP analyses. Results indicate that the co-addition of NS and alkali activator with a water-binder ratio of 0.27 represents the optimal preparation method. The incorporation of NS reduced the 1-day water recovery rate of SLSG from 84.9 % to 40.52 %, thereby extending the geopolymerization period, while still allowing a water recovery rate exceeding 98.5 % after 7 days. The optimal NS dosage was identified as 0.75 wt%, under which the flowability of SLSG measured 175 mm—a 10.26 % reduction compared to the reference group. The 28-day compressive and flexural strengths reached 40.41 MPa and 10.85 MPa, respectively, corresponding to improvements of 40.80 % and 57.02 % over the reference group. The mass-strength efficiencies for compression and flexural performance were 1.78 and 0.48, reflecting increases of 36.92 % and 54.84 % relative to the baseline. SEM and MIP analyses revealed that NS reduced the total porosity of SLSG by 4.52 % through a pore-filling effect, inhibited the formation of coarse pores, sealed through-cracks, and improved crack resistance. XRD, FT-IR, and EDS results confirmed that NS, as an active nanomaterial, promotes geopolymerization and enhances reaction homogeneity via the “nucleation effect” and “chemical bonding effect.” TG-DSC-DTG analyses further demonstrated that NS increases the gel product content and improves high-temperature thermal stability. The densifying and reactivity-enhancing effects of NS improve the applicability of SLSG for lunar surface construction without compromising efficient water utilization, thereby providing theoretical insights and practical references for the development of enhanced lunar construction materials.

* Corresponding author.

E-mail address: lifaping911@whu.edu.cn (F. Li).

1. Introduction

The International Space Exploration Coordination Group (ISECG) aims to explore the Moon's vicinity and surface by 2030 [1]. Following this, NASA launched the Artemis program; the European Space Agency (ESA) released the "Lunar Village" design document; Russia planned the Luna 25–29 series of missions; and the China National Space Administration (CNSA) announced the Lunar Prospecting Phase IV, a major strategic mission, in October 2022 after completing the Lunar Soil Sample Return Mission. A preliminary model for an international lunar research station has been proposed [2]. However, the high cost of Earth-Moon transportation [3] complicates reliance on Earth-based materials for lunar surface construction, and in situ resource utilization (ISRU) technology is expected to address this critical issue [4]. In the short term, the extreme environment of the lunar surface creates challenges to achieve scale utilization of the lunar south pole water ice resources [5]; thus, the construction materials of the lunar research station must withstand the extreme environmental conditions of the lunar surface with a high water utilization efficiency. To address the challenges of constructing in the extreme lunar environment and developing suitable materials, several approaches capable of achieving high mechanical performance have been investigated, including laser sintering [6], microwave sintering [7], and sulfur concrete [8]. However, each of these methods presents certain challenges for practical lunar application. Laser sintering entails a high energy demand, the efficiency of microwave sintering is strongly influenced by the mineralogical composition of the lunar soil simulant, and sulfur concrete has limited high-temperature resistance.

Geopolymers are inorganic cementitious materials synthesized through the alkali activation of silica-aluminum-rich precursors [9], featuring an amorphous three-dimensional network structure. Lunar soil, abundant in silica-aluminum compounds [10], can serve as a precursor for geopolymer formation via alkali activation, offering benefits such as high mechanical strength [11], rapid strength development [12], resilience to extreme temperature cycling [13], and minimal water usage [14]. Furthermore, the geopolymer technique is compatible with 3D printing methodologies [15], enabling highly automated construction on the lunar surface. These combined attributes indicate that geopolymers could be viable in-situ materials for constructing lunar research stations. Davis et al. [16] developed simulated lunar soil-based geopolymer (SLSG) using JSC-1A simulated lunar soil under vacuum and average lunar temperature conditions. Their findings revealed that SLSG cured for 28 d at average lunar temperatures attained a compressive strength of approximately 15.48 MPa, whereas its strength diminished to approximately 3.45 MPa under vacuum conditions. Alexiadis et al. [17] fabricated SLSG using JSC LUNAR-1A simulated lunar soil with a reduced average particle size (19 μm), achieving a compressive strength of 18.4 MPa. This suggests that milling simulated lunar soil to smaller particle sizes can enhance reactivity; however, the milling process may hinder large-scale construction efforts on the lunar surface. Mills et al. [18] prepared and tested SLSG with room-temperature curing (20 °C), low-temperature conservation (−80 °C) and high-temperature exposure (room-temperature conservation followed by 600 °C heating for 1 h) on the compressive strength of three SLSG—BP-1, JSC-1A, and LHS-1. The results showed that heating at 600 °C for 1 h significantly increased the compressive strength of the SLSG, while their compressive strengths were lower than 10 MPa under both room-temperature and low-temperature conditions.

The SLSG prepared in the above studies suffer from low mechanical strengths that are lower than those of the geopolymers prepared with common precursors such as fly ash, slag and other surface common precursors. This may be due to the high crystalline phase of the lunar loam oxides, which leads to their lower reactivity. Consequently, numerous researchers have investigated the reinforcing effects of fibres and inert nanomaterials on SLSG. Debbarma et al. [19] utilized basalt fibres to reinforce LHS-1 and LMS-1 SLSG, assessing their impact on mechanical properties. The results revealed that basalt fibres increased the flexural strength of the material by 32 %, while compressive strength declined by 28 %–63 %. Similarly, Zhang et al. [20] incorporated carbon nanofibers (CNF) into BH-1 SLSG to examine the influence of CNF addition on its mechanical properties. Their findings showed that CNF enhanced flexural strength by 34.8 % and compressive strength by approximately 13.1 %. These results indicate that fibres and inert nanomaterials are inadequate in compensating for the insufficient strength of the lunar surface's primary pressure-bearing components, potentially because they mainly offer physical reinforcement to mitigate material defects. In contrast, reactive nanomaterials exhibit higher reactivity and can directly participate in the geopolymerization process through chemical interactions with alkali activators. Reinforcing SLSG with reactive nanomaterials is expected to address the issue of poor mechanical strength, thereby improving the mechanical properties of geopolymers for more demanding environments and construction requirements.

Nanosilica (Nano-SiO₂, NS), a zero-dimensional reactive nanomaterial characterized by small particle size and high specific surface area [21], has been widely investigated for cementitious material modification. Zidi et al. [22] fabricated metakaolin-based geopolymers incorporating 5 % NS, comparing their mechanical performance under 20 and 80 °C curing regimes. Results indicated superior formation and strength development at elevated temperatures. Deb et al. [23] developed fly ash-based geopolymers with varying NS dosages (1–3 wt%), identifying the 2 wt% NS as the sample with optimal compressive strength, with subsequent strength reduction at higher loadings, suggesting the existence of an optimal NS incorporation threshold. Xu [24] employed a sol-type NS in slag-based geopolymers to evaluate age-dependent mechanical enhancement. NS incorporation increased the compressive strength by 54.5 %, 38.9 %, and 19.0 % after 3, 7, and 28 d of curing, respectively, demonstrating stronger early-stage reinforcement. Additionally, Xu et al. [25] explored the effect of NS particle size on the mechanical properties of geopolymer concrete and discovered that smaller particle sizes provide a more effective filling effect. Revathi et al. assessed NS-modified geopolymers under extreme thermal shock (2 h exposure at 200, 400, 600, and 800 °C). NS incorporation improved their mechanical stability, with reference specimens retaining only 88 % strength at 200 °C, while the NS-modified systems retained 101.8 %–111.4 % strength across the tested dosages.

Collectively, NS can enhance the mechanical properties and improve the thermal stability of geopolymers. Smaller NS particle size and higher curing temperature are more suitable for the development of geopolymer strength, and the effect of NS on the early strength of the material is particularly significant. Given the considerable variations in lunar surface temperature, higher early strength enhances the durability of NS-SLSG under extreme thermal conditions [13]. Consequently, NS enhancement is highly compatible with

lunar construction applications and exhibits substantial potential for extraterrestrial in situ construction. Table 1 details the preparation parameters and compressive strengths of various NS enhanced geopolymers, revealing that geopolymer precursors derive from diverse sources. Variations in particle size, chemical composition, and mineralogical constituents result in differing reactivities, leading to notable differences in optimal NS dosing (0.2–5 wt%) across geopolymer types. Owing to the variability in alkali activator selection, maintenance, and preparation methods, the appropriate preparation techniques, dosages, enhancement effects, and mechanisms of NS-enhanced simulated lunar soil-based geopolymers (NS-SLSG) for lunar surface construction remain unclear and require further investigation.

Therefore, this study utilized Tongji-1 (TJ-1)-simulated lunar soil, which possesses physical and mechanical properties similar to actual lunar soil, as the precursor. Sodium hydroxide (SH), sodium silicate (SS), and calcium hydroxide (CH) were employed as alkali activators to prepare SLSG and incorporate the activated nanomaterial, NS. The effects of NS, the order of the alkali activator doping, and the water-binder ratio on the mechanical properties of NS-SLSG are discussed. Accordingly, NS-SLSG was prepared with different dosages to study the effects of NS on the fluidity, water recovery rate, mechanical properties and mass-strength efficiency of SLSG. Furthermore, a combination of qualitative and quantitative analyses was used to systematically reveal the enhancement mechanism of NS on the mechanical properties of SLSG, aiming to study a building material that is suitable for the construction of the lunar surface with adequately high performance and efficiency in terms of water resource utilization.

2. Materials and methods

2.1. Raw materials

The simulated lunar soil employed in this study was TJ-1, which has been extensively utilized in the Chang'e lunar exploration program and initial sample verification tests for the manned lunar landing program. It is further characterized by mineral composition and physico-mechanical properties similar to those of authentic lunar soil. Table 2 compares the primary chemical compositions of TJ-1-simulated lunar soil with authentic lunar soil samples from the Apollo 14 and Chang'e-5 missions. The comparison in Table 2 demonstrates that the chemical composition of TJ-1 closely mirrors that of real lunar soil, confirming its suitability as a precursor material.

The alkali activator used in the tests consisted of 99 % pure SH (NaOH) in flake form produced by Inner Mongolia Junzheng Chemical Energy Group Co. Ltd. CH $[\text{Ca}(\text{OH})_2]$ and SS ($\text{Na}_2\text{SiO}_3 \cdot 9\text{H}_2\text{O}$) with 95 % purity were used from Tianjin Huasheng Chemical Reagent Co. SS modulus of 1.03 ± 0.03 . NS was selected from VK-Sp30W aqueous dispersion, featuring a particle size of 30 nm and a concentration of 15 %, manufactured by Zhejiang Zhitian Nanomicro New Material Co. Pure water was used as the solvent in the tests. The images of each material are shown in Fig. 1.

2.2. Specimen preparation

The preparation process of NS-SLSG under different mixing orders of NS and alkali activator is illustrated in Fig. 2, and considering

Table 1
Preparation parameters and compressive strength of nano-SiO₂ enhanced geopolymers.

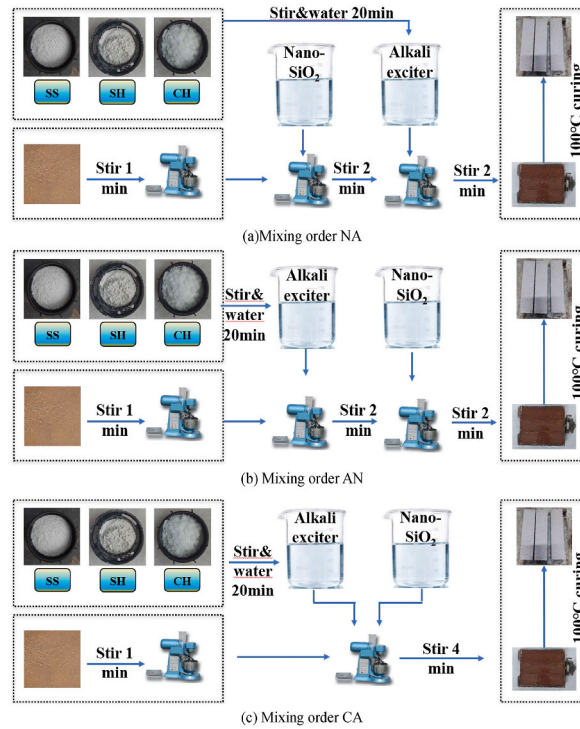
Author	Precursor	Alkali activators	Curing methods	CS (MPa)	Enhancement ratio (%)	Mixing order and dosage
Zidi [22]	Metakaolin	SH, SS	80 °C for 24h + AC	42.32	41.06	PM, 5 wt%
Deb [23]	Fly Ash	SH, SS	AC	About 60	63	PM, 2 wt%
Xu [24]	Slag	SH, SS	40 °C	About 110	19.0	CA, 0.2 wt%
Xu [25]	Coal Gangue + Fly Ash	SH, SS	80 °C +20 °C	44.07	25.27	AN, 1 wt%
Revathi [26]	Fly Ash + Slag	SH, SS	AC	50.4	34.4	AN, 2.5 wt%
Liu [27]	Fly Ash	SH, SS	AC	About 29	11.2	CA, 1 wt%
Luo [28]	Fly Ash	SH, SS	65 °C for 48h + SC	About 35.8	17.38	PM, 2 wt%
Gao [29]	Metakaolin	SH, SS	80 °C	About 68	70	PM, 1 wt%
Durak [30]	Fly Ash	SH	75 °C	36.6	62	NA, 3 wt%
Sun [31]	Metakaolin	SH, SS	AC	About 27	22.7	2 wt%
Rashad [32]	kaolin	SS	50 °C	About 64	10	PM, 0.5 wt%
Khater [33]	Construction and Demolition Waste	SH	WC	About 36	43.3	AN, 2 wt%
Ahmed [34]	Slag	SH, SS	AC	About 37	20.47	NA, 3 wt%
Ibrahim [35]	Volcanic Ash	SH, SS	60 °C	About 42	23.5	PM, 5 wt%
Ibrahim [36]	Volcanic Ash	SH, SS	23 °C	45.0	76.5	PM, 5 wt%
Shi [37]	Sewage Sludge Ash	SH, SS	40 °C	19.7	About 118.9	PM, 2 wt%

Note: Compressive strength (CS). Ambient Curing (AC); Standard Curing (SC); Water Curing at Ambient Temperature (WC). The mixing order points out the different NS and alkali activator mixing orders, which represent Pre-mixed Addition (PM); Co-addition (CA); Alkali Activator First, NS Second (AN); NS First, Alkali Activator Second (NA).

Table 2

Main chemical compositions of TJ-1-simulated lunar soil and Chang'e 5 and Apollo 14 real lunar soils (wt%).

Lunar soil	SiO ₂	Al ₂ O ₃	CaO	MgO	K ₂ O	TiO ₂	P ₂ O ₅	FeO	Na ₂ O	MnO	Ref
TJ-1	47.70	16.20	8.21	5.04	2.29	2.00	0.58	–	4.92	0.15	[38]
Chang'e 5	41.25	11.55	11.64	6.52	0.21	5.12	0.27	22.7	0.46	0.28	[39]
Apollo14	48.10	17.40	10.70	9.40	0.55	1.70	0.51	10.4	0.7	0.14	[40]

**Fig. 1.** Raw materials.**Fig. 2.** Preparation process of NS-SLSG.

the example in Fig. 2(a), the specific implementation process is elaborated.

- (1) Simulated lunar soil was poured into a glue sand mixer and stirred at low speed (140 ± 5 r/min) for 1 min;
- (2) The NS dispersion was injected and mixed at a low speed for 2 min;
- (3) The alkali activator was added with continued mixing for 2 min;
- (4) The geopolymer slurry is injected into the triple test mold ($40 \text{ mm} \times 40 \text{ mm} \times 160 \text{ mm}$), placed on the shaking table and vibrated until no bubbles are observed.
- (5) The mold was placed in the oven at 100°C for 24 h.
- (6) Demolding and continuous curing until 3 d, 7 d, and 28 d before testing.

Early research by Wu [41] indicated that elevated temperatures facilitate the curing of SLSG. Furthermore, due to the Moon's slow rotation, prolonged high-temperature conditions exist in its equatorial and mid-latitude regions [42], making the in-situ preparation of SLSG under lunar surface temperatures feasible. In a related in-situ lunar temperature simulation study, Zhang et al. [43] reported that

a specimen with an initial temperature of 98.2 °C achieved the optimal compressive strength. Consequently, a curing temperature of 100 °C, which is close to this value, was selected for this study.

The mixing order variations are illustrated in steps (2)–(3): NA involves blending the NS dispersion before the alkali activator; AN entails combining the alkali activator prior to the NS dispersion; and CA involves simultaneously mixing the NS dispersion with the alkali activator.

2.3. Experimental design

The alkali activator ratios, based on prior research [44], were prepared using 10 wt% SH, 6 wt% CH, and 6 wt% SS, mixed with pure water for 20 min. The concentration of the NS dispersion was 15 %, and the water-binder ratio was designed considering the total water usage, which should include the water contained in both the NS dispersion and the composite alkali activator. The water-binder ratio was designed considering the total water usage, including that in the NS dispersion and the composite alkali activator.

To investigate the impact of the NS and alkali activator mixing order and the water-binder ratio on the properties of NS-SLSG during preparation, test specimens were fabricated according to the test program presented in Table 3. The optimal preparation parameters for NS-SLSG materials were determined through compressive and flexural strength tests.

Based on the optimal blending order and water-binder ratio determined in Table 3, NS-SLSG with NS blending amounts of 0, 0.50, 0.75, 1.00, and 1.25 wt% were prepared (detailed in Table 4), and the corresponding flowability, water recovery rate, and mechanical properties were systematically tested and simultaneously combined with the analysis of microscopic experiments to reveal the mechanism of the enhancement effect of NS.

2.4. Flowability test

The flow performance of SLSG was assessed following the “Method for Determining the Flowability of Cementitious Sand” (GB/T2419-2005) [45], utilizing the flowability value to quantify flow characteristics. The test employed a shaking table operating at 1 Hz, recording the diameters in two vertical directions along the bottom surface of the slurry after 25 vibrations. The average diameter represented the flowability of the SLSG slurry.

2.5. Water recovery rate measurement

Geopolymer is a low water-demand material [14], with water acting as a mediator in the geopolymerization process and participating minimally in the reaction. Considering the limited water resources on the lunar surface, optimizing water utilization during geopolymer preparation is essential for the sustainability of lunar construction. The water recovery rate at each stage of SLSG was determined using Eqs. (1) and (2).

$$m_{11} = \frac{m_1 - m_2}{m_3} \times m_{12} \quad (1)$$

$$R_n = \frac{m_1 - m_2 - m_n}{m_{11}} \times 100\% \quad (2)$$

Nomenclature: m_{11} : actual water content in SLSG specimen (g); m_1 : post-casting mold mass (g); m_2 : mold tare mass (g); m_3 : total initial material mass (g); m_{12} : designed water content (g); R_n : water recovery efficiency (%); n : curing age (1, 3, 7 d); m_n : specimen mass at designated curing age (g).

2.6. Mechanical property testing

According to the “Test Method for Strength of Cementitious Sand” (GB/T17671-2021) [46], a beam bending and shear loading pressure testing machine was used to perform compressive and flexural strength tests. The flexural strength test utilized a three-point bending setup with a 100 mm span and a loading rate of 50 N/s. Three specimens were evaluated per group, and the flexural strength results were averaged. Thereafter, the fracture block formed by the flexural test was selected for the compressive strength test, with the loading speed set to 2.4 kN/s. Six tests were replicated for each group, and the compressive strength results were averaged.

Table 3
Mixing sequence and water-binder ratio.

Sample	NS doping (wt%)	Water-binder ratio	Preparation sequence
S1	0.75	0.27	NA
S2	0.75	0.27	AN
S3	0.75	0.27	CA
S4	0.75	0.24	CA
S5	0.75	0.30	CA

Note: Doping percentages all represent the mass ratio of the material used to the simulated lunar soil.

Table 4
NS doping.

Sample	Simulated lunar soil	NS	SH	SS	CH	Water
P0	1100	0	110	66	66	297
P0.50	1100	5.50	110	66	66	297
P0.75	1100	8.25	110	66	66	297
P1.00	1100	11.00	110	66	66	297
P1.25	1100	13.75	110	66	66	297

Note: NS values are the actual mass of NS in the dispersion; all units are g.

2.7. Microstructural characterisation

Phase composition was determined using X-ray diffraction (XRD) analysis with a Bruker D8 Advance (Germany) equipped with a Cu K α radiation source ($\lambda = 0.15406$ nm) operated at 40 mA and 40 kV, acquiring diffraction patterns over a 2θ range of 5° – 90° at a scanning rate of $2^\circ/\text{min}$. The chemical structure was characterized by Fourier transform infrared (FT-IR) spectroscopy conducted on a Shimadzu IRTracer 100 (Japan) using the KBr pellet method with a sample-to-KBr ratio of 1:100, scanning wavelengths from 4000 to 400 cm^{-1} on dried powder samples. Thermal properties and the degree of geopolymerization were assessed through thermogravimetric-differential scanning calorimetry-derivative thermogravimetry (TG-DSC-DTG) analyses using a NETZSCH STA449F5 (Germany), spanning a temperature range of 25 – 1000°C at a heating rate of $10^\circ\text{C}/\text{min}$ under a nitrogen atmosphere with a flow rate of $50\text{ mL}/\text{min}$ to prevent oxidation. The pore structure was evaluated via mercury intrusion porosimetry (MIP) tests performed with a Micromeritics AutoPore IV 9520, utilizing a mercury pressure range of 0.5 – $33,000$ PSI in accordance with standard instrument protocols. Micro-morphology was examined using scanning electron microscopy (SEM) analysis on a Zeiss Sigma300 in secondary electron mode at an accelerating voltage of 5 kV , while elemental distribution and geopolymerization homogeneity were characterized through energy dispersive X-ray spectroscopy (EDS) analyses with an Oxford Energy Spectroscopy UL TIMMAX40-Oxford EBSD C-SWIFT system.

3. Results and discussion

3.1. Mixing sequence and water-binder ratio

Fig. 3 illustrates the effects of NS content, alkali activator doping sequence, and water-binder ratio on the mechanical properties of NS-SLGS. The results indicate that the doping order significantly influences material performance, ranking as $\text{CA} > \text{AN} > \text{NA}$ in mechanical strength. Specimens simultaneously doped with NS dispersion and alkali activator (CA group) demonstrated superior mechanical properties, achieving compressive and flexural strengths of 39.69 MPa and 10.77 MPa , respectively. Concurrent mixing can inhibit NS clustering through a highly alkaline environment, while adequate free water facilitates uniform NS distribution, effectively enhancing the material's properties. In contrast, the compressive and flexural strengths of group S1 (NA group) decreased by 19.58% and 38.81% compared to group S3. This decline may result from the initial mixing of NS causing poor dispersion due to insufficient free water and the subsequent addition of the alkali activator failing to break up agglomerates, thereby reducing the reinforcing efficacy of NS. The compressive and flexural strengths of the S2 group (AN group) decreased by 15.22% and 29.25% , respectively, compared with that of S3 group, because the preferential incorporation of an alkali activator would trigger the geopolymerization pre-reaction, which would partially depolymerize the simulated lunar soil particles to form oligomers and even partially form a gel network structure, which would increase the viscosity of the SLGS slurry, and the addition of the NS dispersion in

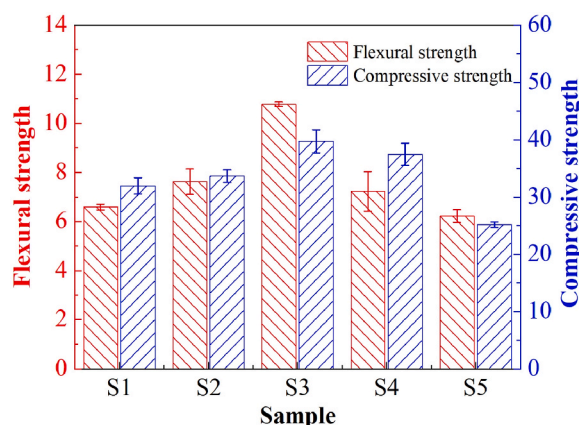


Fig. 3. Mechanical properties of NS-SLGS with different mixing sequences and water-binder ratios.

this case would not be conducive to the homogeneous dispersion of NS, and the formation of a partial gel network might restrict the binding of NS to the interface of the substrate.

A comparative analysis of the S3, S4, and S5 groups reveals that the mechanical properties of NS-SLGS initially increase and then decrease as the water-binder ratio rises. At the optimal ratio of 0.27, adequate free water ensures the uniform distribution of simulated lunar soil particles, thereby minimizing pore defects. Additionally, a moderate alkaline concentration enhances the dissolution of Si and Al in the lunar soil, accelerating oligomer formation and the condensation process. Conversely, at a water-binder ratio of 0.24, slurry fluidity deteriorates, leading to reduced particle packing density, increased bubble retention, and incomplete reactions. Excessive water evaporation during curing at a ratio of 0.30 results in the formation of more capillary pores. Furthermore, diluted alkaline concentrations delay the depolymerization reaction and decrease the amount of depolymerized Si and Al in the early stages, which are crucial for oligomer formation and condensation [29]. Consequently, higher initial water content adversely affects the subsequent geopolymerization reaction. Notably, the higher strength observed at a 0.24 water-binder ratio compared to that at 0.30 further confirms that lunar geopolymer is a low water-demand material and that water plays a minimal role in the alkali activation reaction, serving primarily as a medium to facilitate the polycondensation reaction [44]. Water-binder ratio below 0.24 will lead to poor fluidity and can not be poured and molded, while higher than 0.30 when the phenomenon of water secretion is too serious, so did not test the water-binder ratio below 0.24 and higher than the range of 0.30 NS-SLGS mechanical properties.

In conclusion, the preparation of NS-SLGS specimens with different NS mixing amounts using the CA mixing sequence and a water-binder ratio of 0.27 can be used as the experimental basis for the subsequent study of the effect of NS on the flowability, water recovery rate and mechanical properties of SLGS.

3.2. Water recovery rate

Fig. 4 depicts the water recovery rate trend of SLGS across varying NS dosages. After 7 d of curing at 100 °C, each test group achieved a water recovery rate exceeding 98.5 %, demonstrating that SLGS can attain nearly complete water recovery. This finding highlights the significant benefits of geopolymer molding technology for efficient water resource utilization in the lunar extreme environment and is consistent with the literature [14]. At 1 d, the water recovery rates among the test groups showed significant differences. NS doping decreased the water recovery rate at this stage; as the NS dosage increased from 0.5 wt% to 1.25 wt%, the water recovery rates were 43.73 %, 40.52 %, 47.4 %, and 65.15 %, respectively, all markedly lower than the reference group (P0) at 84.9 %. The P1.25 group exhibited a significantly higher water recovery rate compared to the P0.50-P1.00 groups, likely due to particle agglomeration from excess NS, which weakens the pore-filling effect.

Further analysis showed that the higher water recovery rate (84.9 %) of the reference group at the age of 1 d may form escape channels inside, hindering the densification of the SLGS structure, and the lack of water to act as a reaction medium limits the geopolymerization reaction, leading to a decrease in the gel product and the inability to fill the pore space [47], resulting in the strength of the reference group being lower than that of the NS-SLGS. With extended curing times, the water recovery rate of each component exceeded 96 % after 3 d and surpassed 98.5 % after 7 d. Dehydration and condensation drive the densification of the structure, a key factor in geopolymer strength formation [26]. This indicates that SLGS can achieve rapid strength development within 3 d, demonstrating excellent early-strength characteristics and significant engineering value for the rapid construction of the lunar surface.

The implications of NS doping for potential vacuum applications on the lunar surface warrant further discussion. Studies by Geng et al. [48,49] have identified two distinct setting mechanisms for SLGS: chemical and physical setting. Under atmospheric conditions, the setting process is dominated by chemical setting, where dissolved silicate-aluminate monomers form a strength-giving N-A-S-H gel network. In contrast, under vacuum conditions, rapid water removal creates binding silicate films between particles, which confer some strength but also halt the geopolymerization reaction, resulting in mechanical properties inferior to those obtained through atmospheric curing. The observed reduction in the early water recovery rate due to NS incorporation (Fig. 4) suggests its ability to retain moisture. This moisture-retaining capability could prolong the period available for the geopolymerization reaction in a vacuum

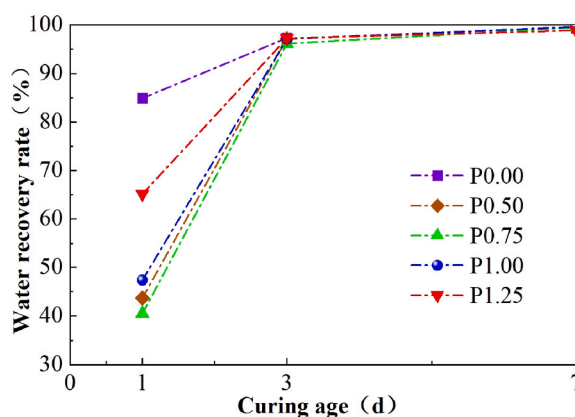


Fig. 4. Water recovery rates of SLGS and NS-SLGS at different curing ages.

before the system irreversibly sets physically. Coupled with its role in accelerating early reaction kinetics, NS incorporation is posited as a viable strategy to optimise the balance between physical and chemical setting, thereby enhancing the final material performance under the vacuum conditions of the lunar surface.

3.3. Flowability test

Fig. 5 illustrates the effect of NS doping on the flow properties of SLSG slurry. As observed in Fig. 5, NS doping reduces the slurry fluidity and gradually decreases the slurry flow properties as the NS doping increases. However, the NS doping reduces slurry fluidity, and the flow properties progressively decrease as NS content increases. The P0 fluidity of the reference group without NS doping was highest, reaching 195 mm. As NS doping increased from 0.5 to 1.25 wt%, the fluidity of each test group decreased to 182, 175, 177.5, and 170 mm, representing reductions of 6.67 %, 10.26 %, 8.97 %, and 12.82 %, respectively, compared to the reference group. This is because of the following reasons: the high surface area of NS particles adsorbs a large amount of free water [50], which reduces the effective amount of water in the slurry system. Moreover, the abundant hydroxyl groups on the surface of NS enhance the interaction between particles through hydrogen bonding [51], which significantly improves the viscosity of the slurry. Although the nanoparticle microaggregate filling effect enhanced slurry compactness at the macroscale, it also increased the particle contact area at the microscale, leading to higher internal frictional resistance.

3.4. Mechanical property

3.4.1. Compressive strength

Fig. 6 shows the effect of NS doping on the compressive strength development of SLSG. NS doping significantly improved the compressive strength of SLSG at all ages, with NS-SLSG strength first increasing and then decreasing as NS doping levels rose. As NS doping was incrementally raised from 0.50 wt% to 1.25 wt%, the 28-day compressive strengths of NS-SLSG attained 38.21, 40.41, 38.62, and 31.51 MPa, corresponding to improvements of 33.14 %, 40.80 %, 34.56 %, and 9.79 % over the reference group (28.7 MPa), respectively. These findings indicate an optimal NS doping threshold. The reduction of NS enhancement effect under high doping may be related to the pore defects caused by nanoparticle agglomeration [52], and a large number of NS agglomerates may lead to the formation of micrometer-sized crystals, which seriously reduces the nanoeffect of NS and weakens its enhancement effect. As such, NS can be synergistically enhanced to increase its compressive strength through multiple reinforcement mechanisms, which are mainly reflected in the nano-enhancement mechanism and the active enhancement mechanism.

The nano-enhancement mechanism is predominantly exhibited through three key aspects: (1) the highly active surface of NS enhances the kinetics of the geopolymerization reaction [53], facilitating the cleavage of Si-O-Si/Al bonds in simulated lunar loam and releasing silica-aluminum monomers $[\text{Si}(\text{OH})_4, \text{Al}(\text{OH})_4]$; (2) the nanoparticles act as nucleation sites, promoting the condensation of silica-aluminum monomers with Na^+ and Ca^{2+} to form N-A-S-H and C-(A)-S-H gel networks, thereby increasing the overall number of gel phases; and (3) the nanoscale pore-filling effect decreases defect density within the material, improves structural density, and enhances the uniformity of stress distribution.

The primary mechanism for activity enhancement involves NS reacting with alkali activators and directly participating in the geopolymerization process, forming ionic bonds and stable Si-O-Si bonds with geopolymeric gel products [51]. This introduces a distinct enhancement mechanism, namely the chemical bonding effect. Zhu [54] demonstrated that under alkaline conditions, the dissolution of raw materials is selective, with Al sites dissolving more rapidly than Si sites. This selective dissolution results in a swift accumulation of Al sources during the initial stages, whereas the limited Si sources significantly inhibit early geopolymerization. Unlike the free $\text{Si}(\text{OH})_4$ derived from the depolymerization of raw materials, NS has certain structural properties that are internally connected by Si-O-Si bonds and carries hydroxyl groups on the surface that can attract $(\text{Al}(\text{OH})_4)$ dehydroxylation to produce

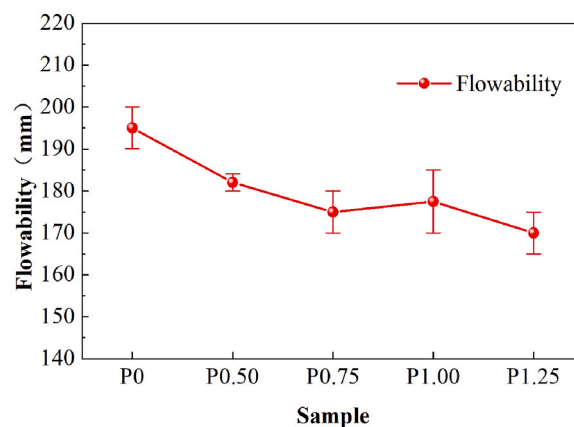


Fig. 5. Fluidity results.

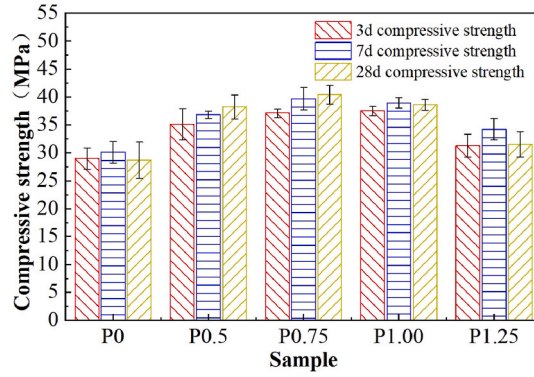


Fig. 6. Compressive strength test results.

polycondensation (Eq. (3) for the principle of dehydroxylation condensation of NS). NS undergoes dehydroxylation condensation, which embeds nanoparticles into the three-dimensional gel network (refer to Fig. 7 for a schematic of the chemical bonding effect of NS). This process indicates that NS can participate in condensation reactions without requiring depolymerization or oligomer formation, thereby directly forming a more structurally robust gel network. Consequently, NS significantly enhances the geopolymerization reaction and balances precursor depolymerization and polycondensation. Together, these mechanisms lead to the densification of the SLSG microstructure.



Further analysis shows that both SLSG and NS-SLSG exhibit substantial early strength, aligning with the findings in Section 3.2 and suggesting a strong correlation between moisture loss and mechanical property development. Specifically, the P0.75 group achieved 91.83 % and 98.22 % of their 28 d strengths at 3 and 7 d, respectively. This rapid strength development of NS-SLSG effectively meets the requirement for high early-age strength necessary for constructing the moon table. This result is consistent with the role of high-temperature curing in accelerating the early reaction process and concurs with the conclusions of Gunasekara et al. [55]. Note that certain specimens showed a strength inversion phenomenon (4.76 % reduction in strength in the P0 group and 7.89 % reduction in the P1.25 group) after 28 d, which may be related to the dehydration and decomposition of part of the gel phase triggered by the prolonged high temperature [30].

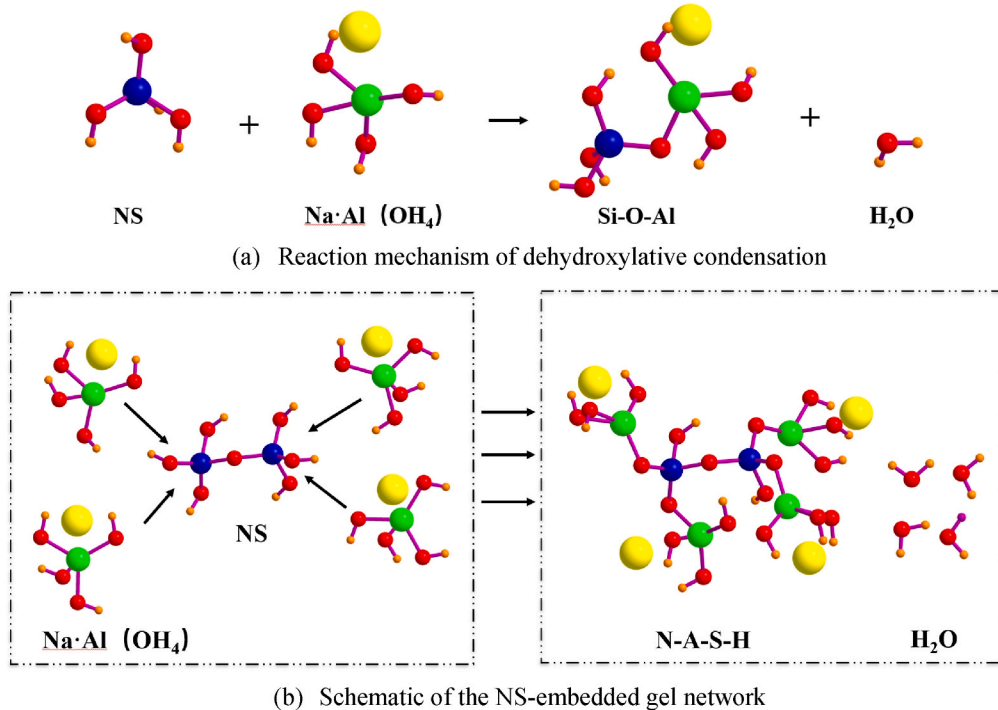


Fig. 7. Schematic of the NS chemical bonding.

A moderate NS doping level (0.75 wt%) effectively enhances the thermal stability of the material. NS particles establish strong chemical bonds with silica-oxygen tetrahedra (SiO_4) and aluminum-oxygen tetrahedra (AlO_4) within the gel network, thereby reinforcing the structure, preventing the propagation of high-temperature microcracks, and preserving structural integrity. However, excessive NS doping hinders the incorporation of NS into the gel network, resulting in a swift reduction of the enhancement effect beyond the optimal threshold. This decline is attributed to agglomeration, which increases NS particle size and diminishes the number of surface hydroxyl groups, thereby weakening the chemical bonding.

3.4.2. Flexural strength

Fig. 8 illustrates the impact of NS doping on the flexural strength of SLSG. NS doping markedly improves the flexural strength of SLSG, with the flexural strength of NS-SLSG first increasing and then decreasing as NS doping levels rise. When the NS doping was gradually increased from 0.50 to 1.25 wt%, the 28-day flexural strength of NS-SLSG reached 10.29, 10.85, 9.74, and 7.72 MPa, respectively, corresponding to an enhancement of 48.91 %, 57.02 %, 40.96 %, and 11.72 % compared with that of the reference group. Thus, the enhancement of flexural strength of NS is similar to that of compressive strength. The addition of NS similarly enhances both flexural and compressive strengths, with an optimal doping level identified at 0.75 wt%. The flexural strength exhibits excellent early strength, and the 3- and 7-day flexural strengths of the P0.75 group can reach 82.58 % and 99.26 % of the 28-day strength, which, combined with the development law of the compressive strength, confirms the early strength characteristics of NS-SLSG.

It is noteworthy that the enhancement effect of NS on the flexural strength of SLSG is more pronounced than on its compressive strength. However, the strength deterioration induced by excessive NS dosage is also more significant in flexural tests. Specifically, the 28-day flexural strength of the P1.25 group decreased by up to 28.85 % compared to the P0.75 group. This can be attributed to the tensile-compressive composite stress field the material is subjected to during the three-point bending test, which exhibits higher sensitivity to the development and propagation of cracks under load, as well as to the material's fracture toughness.

Golewski et al. [56,57]. employed Digital Image Correlation (DIC) technology to investigate the effect of NS incorporation on the fracture toughness and crack development trends of materials under three-point bending load. The results indicated that NS incorporation increased the material's fracture toughness by 19 %, resulting in higher fracture mechanics parameters. Furthermore, while NS incorporation did not significantly alter the material's failure mode, which remained characteristically brittle, observable crack development could be detected prior to failure. Compared to the reference group, the NS-incorporated group exhibited increased crack width. Due to its brittle failure characteristics, the fracture failure of SLSG is generally sensitive to inherent internal defects.

Combining the MIP analysis results from Section 3.5.4 and the SEM analysis from Section 3.5.5, it is understood that an appropriate amount of NS fills pores through the physical filling effect, reduces the porosity of SLSG, and inhibits the formation of coarse pores larger than 300 nm. Moreover, NS can seal microcracks and suppress the formation of through-cracks, thereby reducing the density of inherent defects. The fracture failure in the P0 group is primarily due to the propagation and extension of internal large cracks and coarse pores towards weaker areas under load. The connection and combination of these internal defects lead to the formation of through-cracks, resulting in its lower flexural strength. In contrast, for NS-incorporated SLSG, due to its higher internal density, fracture failure requires the consumption of more energy, thereby significantly improving the flexural strength. Therefore, the increase in the number of coarse pores in SLSG caused by excessive NS introduces certain structural defects, which consequently weakens the crack resistance of the material.

3.4.3. Mass-strength efficiency

Utilizing the mass-strength efficiency (MSE) evaluation system proposed by Zhou et al. [58], this study systematically evaluated the optimization of NS resource utilization in SLSG. It responds to the mass of admixture required for the material to form a unit strength (1 MPa), and the larger value indicates that less mass of admixture is required, and the calculation formula is shown in Eq. (4).

$$\eta_{C/Fm} = \left[\frac{(m_{AE} + m_{NS}) \times 100}{m_{TJ} \times f} \right]^{-1} \quad (4)$$

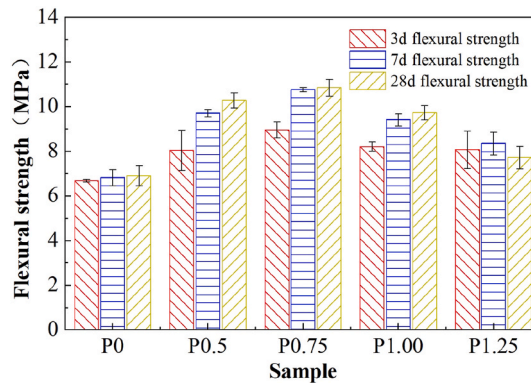


Fig. 8. Flexural strength test results.

Nomenclature: m_{AE} : Quality of alkali activators (g); m_{NS} : Quality of NS (g); m_{TJ} : Quality of TJ-1-simulated lunar soil (g); f : Compressive or flexural strength (MPa); η_{Cm} , η_{Fm} : Compressive and flexural strength mass strength efficiency.

Fig. 9 displays the mass strength efficiency for compressive and flexural strengths of each test group at 28 d. NS doping notably improved the SLSG at this time point, with the P0.75 group achieving a value of 1.78, marking a 36.92 % increase over the reference group. This reduction suggests that the mass of earth replenishment material required to achieve a compressive strength of 1 MPa in lunar soil was decreased by 36.92 % compared to the reference group. The curves show the same optimal doping level (0.75 wt%) as observed in mechanical strength development, and the increase drops to 4.62 % when NS doping reaches 1.25 wt%, consistent with performance decline due to nanoparticle agglomeration.

NS exerts a stronger optimization effect on bending performance, as the P0.75 group (0.48) is 54.84 % higher than the P0 group, demonstrating that NS significantly enhances the cost-effectiveness of the material's flexural performance. When the NS dosage reaches 1.25 wt%, the increase decreases to 6.45 %, indicating that the economic optimization effect of NS on flexural strength parallels that on compressive strength, and an optimal threshold exists. This phenomenon reveals a strong coupling relationship between the material performance-economy indexes, i.e., a moderate amount of NS (0.75 wt%) can realize the double enhancement of mechanical properties and resource efficiency through the synergistic effect of pore-filling-gel enhancement. In contrast, the agglomeration defects caused by excessive doping of NS will weaken the benefits of both.

Combining the results from Sections 3.4.1 and 3.4.2 reveals that doping SLSG with an appropriate amount of NS significantly reduces dependence on Earth resources while maintaining superior mechanical properties. This confirms the potential of NS to lower material costs. The performance-cost synergistic optimization is expected to provide a sustainable material solution for resource-constrained applications, such as lunar surface construction. Its core value lies in (1) reducing Earth recharge mass by enhancing the utilization efficiency of in situ lunar soil and (2) optimizing the gel reaction process to decrease alkali activators consumption.

3.5. Microstructure

3.5.1. XRD analysis

Fig. 10 presents the XRD test results for the P0, P0.50, P0.75, and P1.25 groups. It shows that the primary crystalline phases in SLSG include olivine (Mg_2SiO_4 PDF No 87–0677) and albite-high ($Na(AlSi_3O_8)$, PDF No. 99-0001), among others, which is in agreement with findings reported in other SLSG studies [59,60]. Furthermore, all test groups exhibited similar phase compositions, indicating that NS doping did not alter the type of geopolymerization products. The diffraction pattern has a large area of a weak hump in the 20° – 36° (2θ) interval, and this amorphous peak may originate from the precursor products of C-(A)-S-H and N-A-S-H gels [61], i.e., a large number of amorphous gels were produced by SLSG and NS-SLSG during alkali excitation [62]. The amorphous peak area increased proportionally with the amorphous material content. The NS-SLSG group exhibited a larger or comparable area under the amorphous peak compared to the reference group, demonstrating higher peak intensity. This indicates that NS doping enhances the geopolymerization reaction.

Fig. 10(b) displays a broad hump at approximately 29° – 30° (2θ), signifying the formation of N-A-S-H and C-(A)-S-H gels [63,64]. The intensity of this characteristic peak decreased in the order $P0.75 > P0.50 > P1.25 > P0$. A slight peak shift was observed, attributed to the mutual improvement effect [22]. The P1.25 group showed a prominent quartz characteristic peak at 68° (2θ) (PDF No. 83–5039), likely resulting from NS particle agglomeration that formed micrometer-sized crystals exhibiting diffraction behavior similar to bulk SiO_2 . This further confirms that excess NS induces agglomeration phenomena, consistent with the deterioration of the mechanical properties in the P1.25 group.

Further analysis of Fig. 10(a) shows that the alkaline environment drives the production of zeolite-like minerals (sodalite ($Na_8Al_6Si_6O_{24}(OH)_2(H_2O)$, PDF No. 76–1639), cancrinite ($Na_8(Al_6Si_6O_{24})(OH)_{2.04}(H_2O)_{2.66}$, PDF No. 98–0147)), and the high-temperature phase of cristobalite (SiO_2 , PDF No. 076–0941). This observation is in agreement with earlier studies [43,65], thus confirming that zeolitic phases can form under such high-temperature and alkaline conditions. The formation of these aluminum-rich

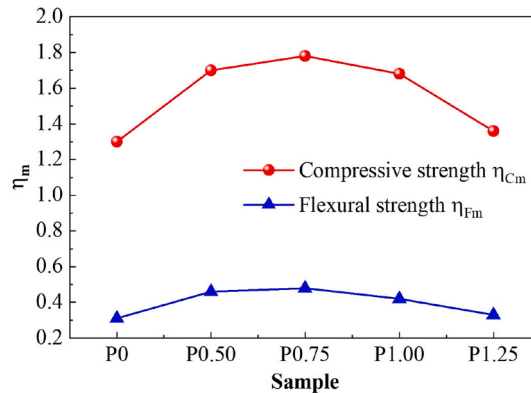


Fig. 9. Mass-strength efficiency results.

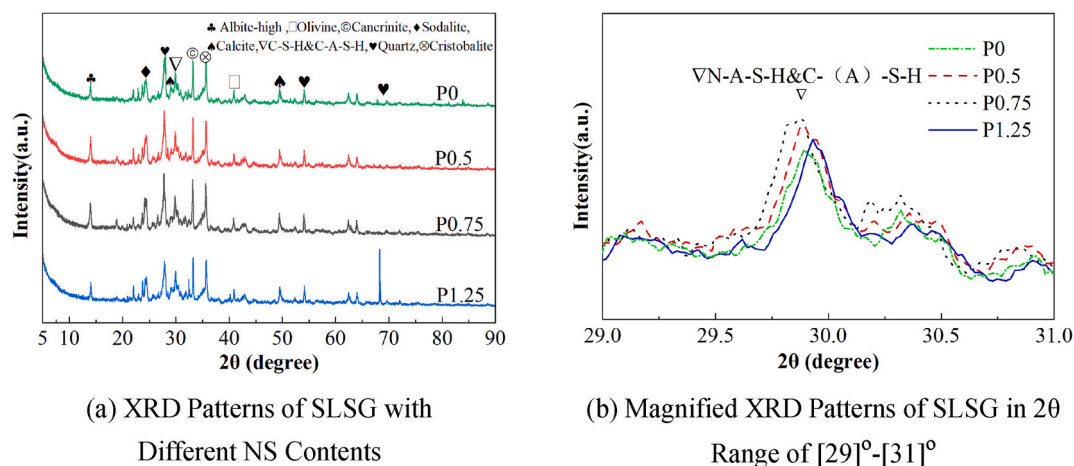


Fig. 10. XRD patterns of SLSG.

zeolitic phases is primarily due to the reaction of Na^+ ions with locally formed silicate oligomers and aluminum monomers, leading to their crystallization and precipitation. Calcite diffraction peaks (PDF No. 33–0268) were identified at 30° and 49° (2θ), consistent with subsequent FT-IR results, thereby confirming slight carbonation of the material. Given the limitations of XRD for quantitatively analyzing amorphous phases, the phase-specific thermal decomposition behavior will be investigated using the TG-DSC-DTG technique to obtain more comprehensive microstructural information.

3.5.2. FT-IR analysis

Fig. 11 illustrates the FT-IR spectra for the P0, P0.50, P0.75, and P1.25 samples. The predominant absorption band occurs at 988 cm^{-1} , corresponding to the asymmetric telescoping vibration of Si-O-T (where T = Si/Al) [22,66,67]. Peak intensity decreases in the order $\text{P1.25} > \text{P0.75} > \text{P0.50} > \text{P0}$, indicating that NS doping significantly enhances gel phase formation. However, despite the P1.25 group exhibiting the highest peak intensity, its mechanical properties were relatively compromised. XRD analysis results (refer to Section 3.5.1) confirm that this decline is due to interference from Si-O-Si bending vibrations caused by excessive NS agglomeration.

Further analysis demonstrates that the absorption peaks in the interval $600\text{--}800 \text{ cm}^{-1}$ correspond to the Si-O-Al bending vibration [29,53], indicating that the simulated lunar soil particles were excited by alkali to release the silica-aluminum monomers and condensed to form an aluminum-containing gel, i.e., confirming the formation of the C-A-S-H and N-A-S-H gel; the characteristic quartz peak at 735 cm^{-1} reflects the presence of silicate minerals. The absorption peak at 1457 cm^{-1} originates from the asymmetric stretching mode of O-C-O bonding, which may be due to the slight carbonation caused by partial CO_2 intrusion from the air [62]; and the difference in bending vibrational intensities of the hydroxyl peaks at 1641 and 3437 cm^{-1} exhibited that NS doping elevated the retention of bound water, which was conducive to the enhancement of gel network stability.

3.5.3. TG-DSC-DTG analysis

Fig. 12 illustrates the TG-DSC-DTG test results for the P0, P0.50, P0.75, and P1.25 groups. These groups display similar TG-DSC-DTG curves, suggesting comparable internal structures and reaction products. This indicates that NS does not affect the reaction type of

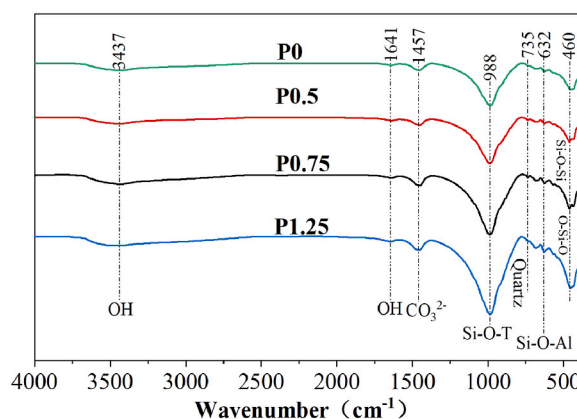


Fig. 11. FT-IR spectra of SLSG and NS-SLSG.

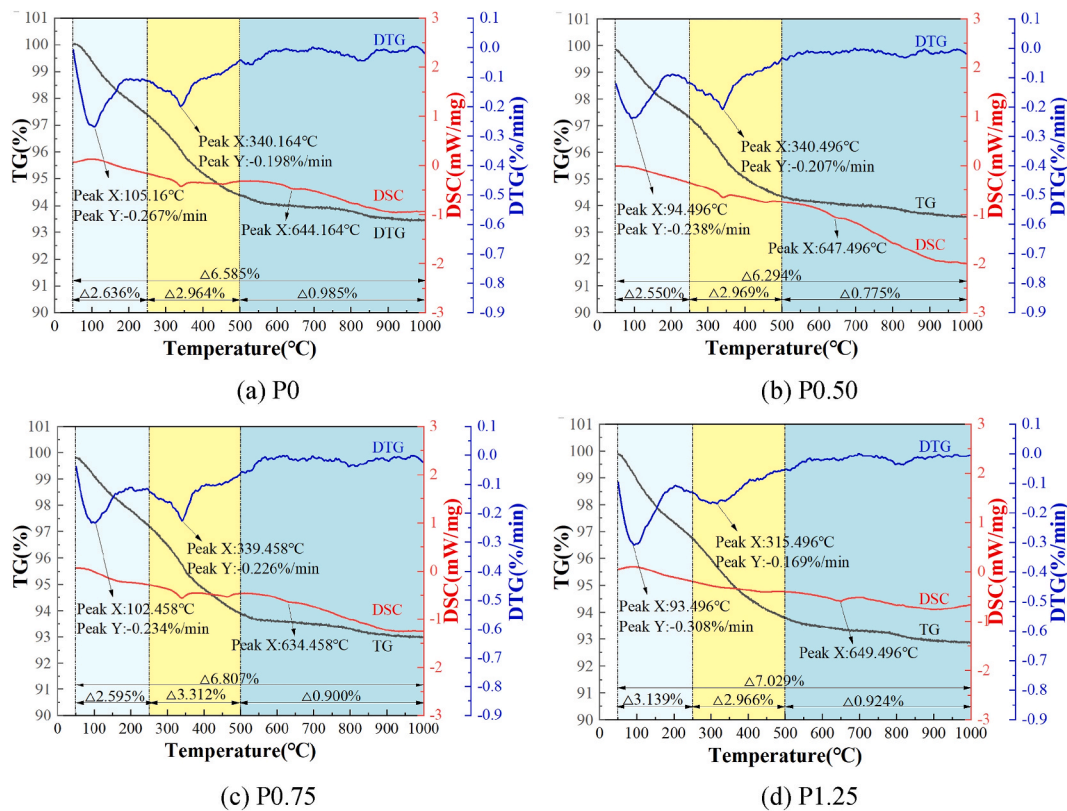


Fig. 12. TG-DSC-DTG curves of SLGS and NS-SLGS.

SLGS, aligning with the crystalline phase detection results from the XRD analysis in Section 3.5.1. Furthermore, the total mass loss increases with higher NS doping, consistent with the findings of Zidi et al. [22]. Based on the thermal decomposition characteristics, the temperature interval can be segmented into three characteristic regions:

The mass loss in region A (50–250 °C) mainly originated from the evaporation of free water and the desorption of some weakly bound water [68,69], and the results of the mass loss rate were in the group of $P1.25 > P0 > P0.75 > P0.50$, with the mass loss rate in the group of P1.25 being significantly higher than that of the remaining three groups. This, combined with the observation that the DSC curve did not produce a significant heat absorption peak in the region and the results of FT-IR detection in Section 3.5.2 results of the hydroxyl diffraction peak, confirmed that the excess NS (1.25 wt%) adsorbed more water, resulting in a significant increase in mass loss. The DTG curve produced the largest peak in region A, indicating that the rate of mass loss in region A was higher than that in other regions, and the rate of mass loss in the P0, P0.50, and P0.75 groups was close to that of approximately 0.234–0.267 %/min, and similarly the rate in the region. The rate of mass loss in group P1.25 (0.308 %/min) was significantly higher than that of the other experimental groups, which is consistent with the maximum rate of mass loss produced in region A. Indicatively, free and weakly bound water are more sensitive to temperature changes in region A [64].

The mass loss observed in region B (250–500 °C) is due to the loss of chemically bound water and the dehydroxylation of geopolymer gels [70–72]. At these temperatures, most gels undergo condensation until structural damage occurs, as demonstrated by DTG curves showing a considerably higher mass loss rate and DSC curves exhibiting a significant heat absorption peak in this region. The mass loss rate in region B follows the order $P0.75 > P0.50 > P1.25 > P0$. Notably, the P0.75 group experienced an 11.74 % increase in mass loss rate compared to the P0 group. This result indicates a higher content of hydroxyl groups and chemically bound water, and demonstrates that the incorporation of NS enhances the degree of geopolymerization in SLGS. [73]. However, the mass loss rates of the P0.50 and P1.25 groups were only slightly higher than those of the P0 group, suggesting that NS has a limited capacity to enhance the geopolymerization reaction at either lower or excessive dosages.

The mass loss in region C (500–1000 °C) tends to stabilize, and the mass loss in this region may originate from the viscous sintering and vitrification reaction under the action of high temperatures [74], and this change may cause the decomposition of the main structure, and its mass loss rate shows the pattern of $P0 > P1.25 > P0.75 > P0.50$, and the P0.75 group reduces the high-temperature mass loss rate by 8.6 % compared with the P0 group. The results confirmed that NS improves the thermal stability of SLGS at elevated temperatures. Both DTG and DSC curves exhibited a minor peak at approximately 650 °C, likely due to the further decomposition of carbonate rock formed by the geopolymer gel [75], leading to increased mass loss and an endothermic reaction. At approximately 800 °C, a broad peak appeared in the DTG curves of all samples, primarily resulting from crystallization reactions [76] associated with calcite decomposition.

Comprehensive analysis demonstrated that NS significantly enhanced the mechanical properties and thermal stability of SLSG by promoting gel phase formation (11.74 % increase in mass loss in region B) and optimizing high-temperature stability (8.6 % decrease in mass loss in region C). This synergistic enhancement of thermal stability and mechanical properties ensures the applicability of SLSG in extreme lunar surface environments.

3.5.4. MIP analysis

Fig. 13 displays the cumulative pore volume distribution curves for the P0, P0.50, P0.75, and P1.25 groups. According to the pore size classification criteria [77], the internal pore size of SLSG primarily spans in the capillary pore range of 100–300 nm, which originates from the evaporation of free water. This finding is consistent with the results of the water recovery rate, i.e., drastic water evaporation under the high-temperature conditioning is the key factor of the pore composition of SLSG. Further analysis of Fig. 13 revealed that the reference group (P0) exhibited a characteristic peak at 9286 nm within the coarse pore region (>300 nm). In contrast, the P0.50 and P0.75 groups did not display significant peaks in this area, suggesting that a moderate amount of NS inhibits the formation of coarse pores. This result is consistent with the findings of She et al. [78], and is attributed to the physical filling effect of NS at the nanoscale, whereby the nanoparticles fill the pores and suppress their initiation, thereby significantly reducing the coarse pore content. In contrast, the excessive NS group (P1.25) exhibited characteristic peaks at 3942 and 6038 nm in the coarse pore region. This observation, which is consistent with the conclusion of Xi et al. [79], indicates that overdosing NS compromises its physical filling efficiency due to agglomeration. Consequently, this promotes the re-formation of coarse pores and adversely affects the overall pore structure, ultimately leading to the reduction in mechanical performance observed in group P1.25 compared to group P0.75. This finding is consistent with the flexural strength results presented in Section 3.4.2. Additionally, although the coarse pore size in the P1.25 group decreased compared to the P0 group, suggesting a reduction in the physical enhancement effect due to over doping, it still represents an overall improvement over the reference group.

Table 5 displays the pore structure parameters for the P0, P0.50, P0.75, and P1.25 groups. From Tables 5 and it can be seen that the total porosity of NS-SLSG increases with the increase of the dosage of NS. However, they all significantly decreased compared with the P0 group. Compared with the reference group, P0 (40.09 %), the porosity of the P0.50, P0.75, and P1.25 groups decreased by 4.88 %, 4.52 %, and 2.67 %, respectively. In addition, NS displays an improvement pattern similar to that of porosity on the bulk density and total pore volume of SLSG.

The P0.50 group demonstrates the lowest porosity; however, the TG-DSC-DTG analysis in section 3.5.3 reveals that low-doped NS has a limited ability to facilitate the geopolymerization reaction. This indicates that low-doped NS primarily enhances the pore structure through physical filling, thereby improving performance. Although the P0.75 group possesses slightly higher porosity than the P0.50 group, its porosity remains 4.52 % lower than that of the P0 group. Combined with the findings in section 3.5.3, this suggests that an optimal NS dosage simultaneously offers effective physical-chemical synergistic enhancement, evidenced by an 11.74 % increase in gel content and a 4.52 % reduction in total porosity. In contrast, the P1.25 group shows increased porosity due to agglomeration effects; nonetheless, it still outperforms the reference group. This observation is consistent with the suppression of coarse pores resulting from NS overdoping.

The analysis of pore size parameters revealed that NS doping decreased the mean, median, and most probable pore size of SLSG. NS doping decreased the mean pore sizes of the P0.50, P0.75, and P1.25 groups by 27.84, 4.30, and 8.62 nm, respectively, compared with that of the reference group (241.02 nm). In addition, the most probable and median pore sizes of the P1.25 group were lower than those of the P0.50 and P0.75 groups, which may be attributable to the increased proportion of gel pores (<10 nm) in P1.25 groups. Nonetheless, the elevated total porosity and larger coarse pore sizes still led to reduced strength.

3.5.5. SEM analysis

Fig. 14 displays the SEM micrographs of the P0, P0.50, P0.75, and P1.25 groups at magnifications of $\times 2000$ and $\times 500$. As shown in

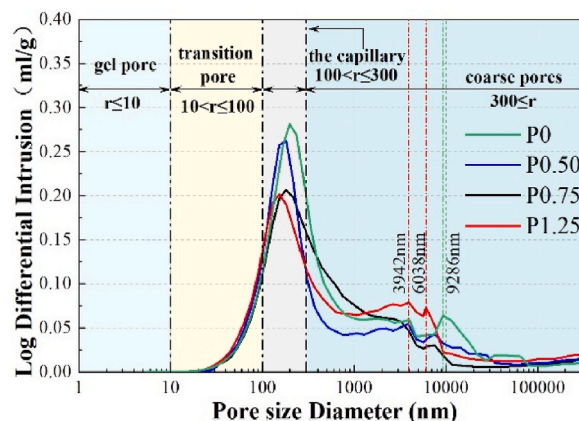


Fig. 13. Cumulative pore volume distribution curves of specimens with SLSG and NS-SLSG.

Table 5

Characteristic parameters of the pore structure of SLSG and NS-SLSG.

Sample	Porosity (%)	Hole capacity (mL/g)	Bulk density (g/mL)	Average pore size (nm)	Most probable pore size (nm)	Median pore size (nm)
P0	40.09	0.2593	1.5462	241.02	198.90	143.55
P0.50	35.21	0.2173	1.6203	213.18	182.31	129.37
P0.75	35.57	0.2240	1.5883	236.76	181.65	137.04
P1.25	37.42	0.2387	1.5675	232.40	153.15	117.64

Fig. 14(a), the reference group (P0) demonstrated a typically loose structure with numerous wide cracks, pores, and unreacted simulated lunar soil particles. The geopolymerized gel product predominantly formed a weak gel layer, which retained a substantial number of pores and cracks. This porous, weak gel layer structure significantly decreased the effective bearing area of the material, thereby weakening its mechanical properties. Combined with the test results discussed in Sections 3.3 and 3.4, the reason may be that the high water recovery rate (84.90 %) in the early stage created a large capillary stress, which resulted in the emergence of pores and microcracks. The lack of nucleation sites restricts the cross-linking of the silica-aluminum precursor, which affects the formation of a gel network that is unable to build up a continuous and homogeneous three-dimensional skeleton structure. Consequently, the gel network displays fractures, voids, and an irregular distribution. Furthermore, the research by Golewski et al. [80,81], on crack development and the interfacial transition zone (ITZ) demonstrates that larger crack widths and a pronounced ITZ in the microstructure can significantly reduce the fracture toughness of materials. These initial microcracks act as inherent defects, which propagate rapidly under external stress, leading to failure and a consequent reduction in mechanical properties. This mechanism is consistent with the flexural strength analysis presented in Section 3.4.2.

As illustrated in Fig. 14 (b, c, d), NS doping markedly improves the microstructure of SLSG, with a dense gel layer dominating the interior of NS-SLSG. In the P0.50 group, only minor pores and cracks are observed, suggesting that low levels of NS doping can fill pores, seal microcracks, and inhibit crack propagation. This enhances mechanical properties and provides robust physical reinforcement. These results are consistent with Huo's study [53] and the MIP test outcomes detailed in Section 3.5.4. However, the SLSG surface remains rough and contains more unreacted simulated lunar soil particles, indicating that low NS dosage does not sufficiently advance the geopolymerization reaction and restricts chemical enhancement. This is in agreement with the findings from the FT-IR, XRD, and TG-DSC-DTG analyses.

As observed in Fig. 14(c), the P0.75 group exhibits a well-developed gel structure with no visible wide cracks and only limited microcracks and pores, which can be attributed to the gel pores formed during the construction of the three-dimensional gel network via geopolymerization. The failure of SLSG is generally highly correlated with the presence of wide cracks and coarse pores. The influence of these inherent defects on the mechanical properties of SLSG is significantly more pronounced than that of microcracks and gel pores. Therefore, although the P0.75 group contains a small number of microcracks, its mechanical properties are still markedly improved compared to the reference group. The internal structure of the P0.75 group is smooth, homogeneous, and dense. As it is difficult to observe unreacted simulated lunar soil particles, NS provides both physical and chemical enhancements under moderate doping.

Fig. 14(d) demonstrates that the P1.25 group exhibits a higher number of cracks and pores compared to the P0.50 and P0.75 groups. Nevertheless, the pore volume is markedly reduced relative to the P0 group, indicating that increased NS doping still enhances the pore structure of the SLSG. The gel structure of the P1.25 group is primarily composed of a semi-dense gel layer, a distinctive feature likely arising from three factors: (1) In the P0 group, the early dehydration rate does not adequately match the crosslinking rate of silicate precursors during gel formation, resulting in a nonhomogeneous gel. Similarly, in the P1.25 group, the water removal rate fails to synchronize with the crosslinking rate of the silica-aluminate precursor during gel formation, leading to a nonhomogeneous structure and localized semi-dense regions. (2) Intermediate state porosity may also be a critical factor, the pore distribution within the gel occupies interstitial spaces, thereby partially compromising the physical properties of the gel phase. (3) The agglomeration of NS may hinder the complete densification of the gel structure, thus forming a semi-dense structure.

Based on the experimental results from Sections 3.5.4 and 3.5.5, a suitable NS content (0.75 wt%) facilitated the complete cross-linking of the silica-aluminate precursor via the nucleation effect. This, combined with nano-filling, optimized the pore structure and ultimately resulted in a dense, continuous composite structure.

3.5.6. EDS analysis

Fig. 15(a–d) shows the EDS results for different dense regions (Area 1 and Area 2), with sampling points marked in the SEM images at 500 × magnification (Fig. 15). Table 6 presents the atomic number ratios of each element along with their standard deviations (SD) for each test group across different areas, enabling the assessment of the geopolymerization reaction degree and the homogeneity of the SLSG [62]. Fig. 15 and Table 6 indicate that the Ca/Si ratios in all test groups are generally below those of C-S-H gels (0.8–1.5 [82]), while the Al/Si ratios are elevated, suggesting that SLSG primarily consists of C-A-S-H and N-A-S-H gels. Further analysis reveals that incorporating NS significantly improved the reaction homogeneity, and the standard deviations of Al/Si, Na/Al, and Ca/Si ratios in each region of the P0.75 group (SD = 0.04, 0.12, and 0.01) were reduced by 76.47, 68.42, and 91.67 %, respectively, compared with the reference group (SD = 0.17, 0.38, and 0.12). This finding suggests that the ground of the P0.75 group polymerization reaction is more homogeneous, i.e., NS promotes the spatial homogenization of the geopolymerization reaction, which is consistent with the findings of Yang [62] et al.

Table 6 demonstrates that the reference group (P0), despite exhibiting higher Na/Al ratios (0.74–1.50), experienced

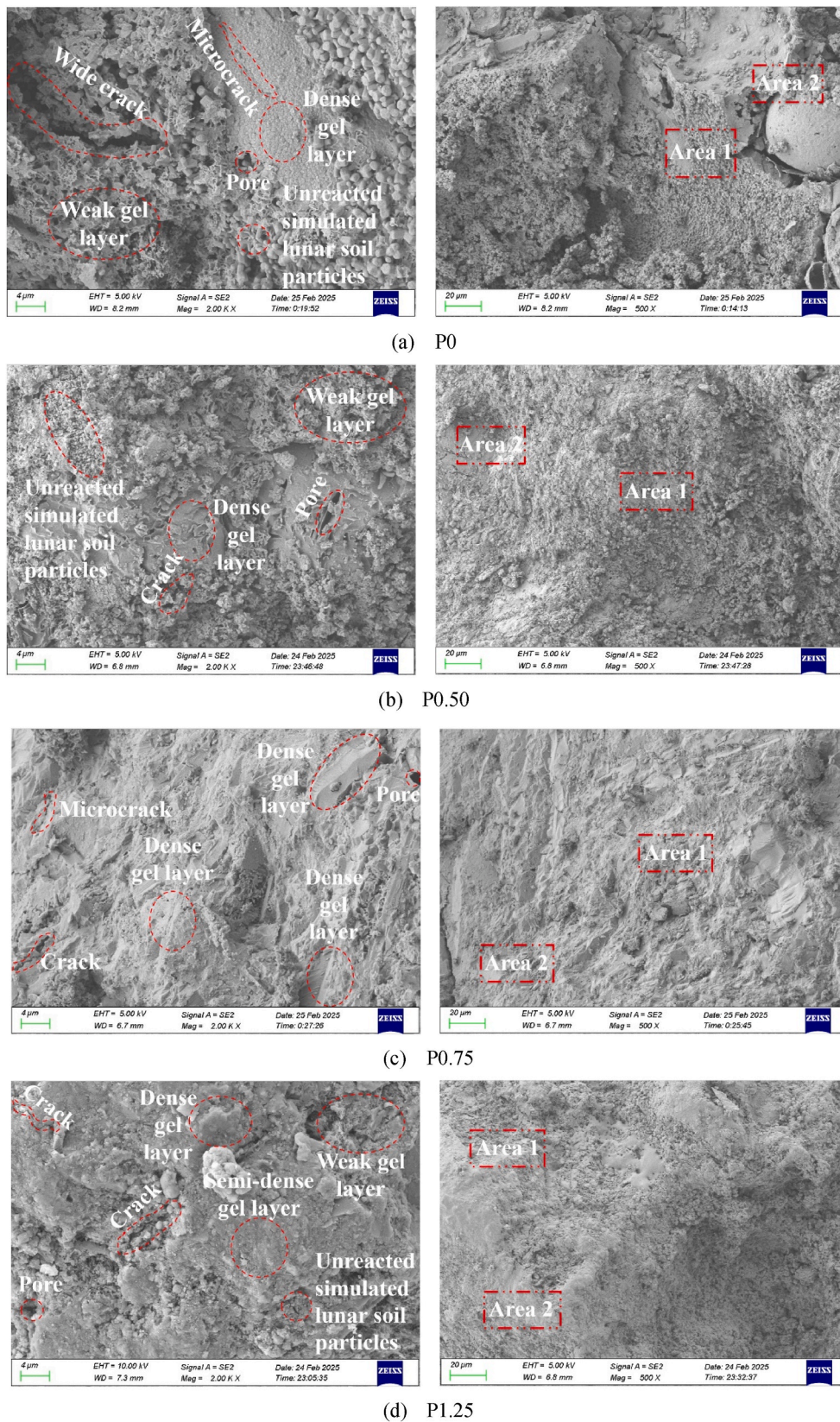
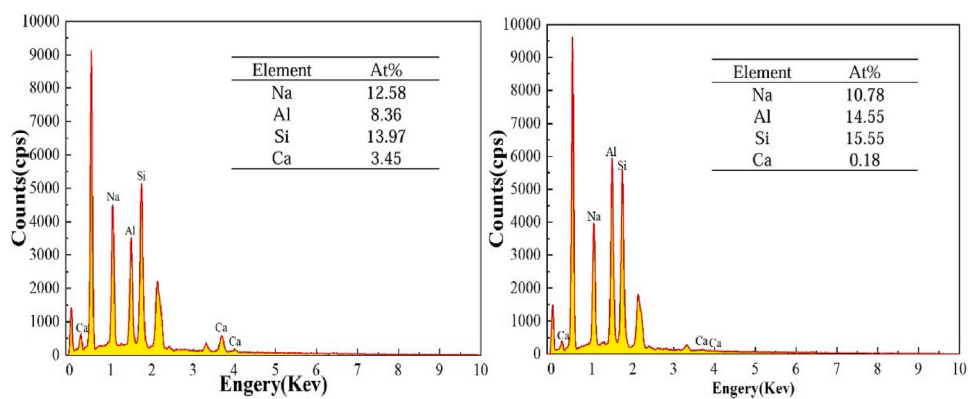
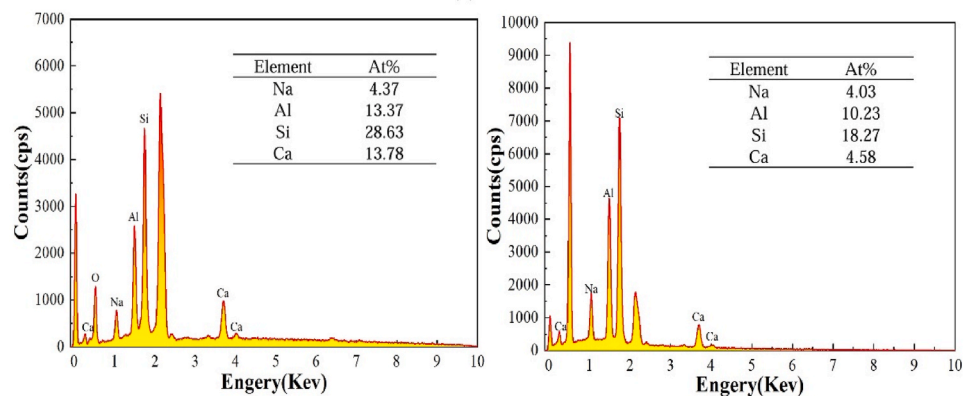


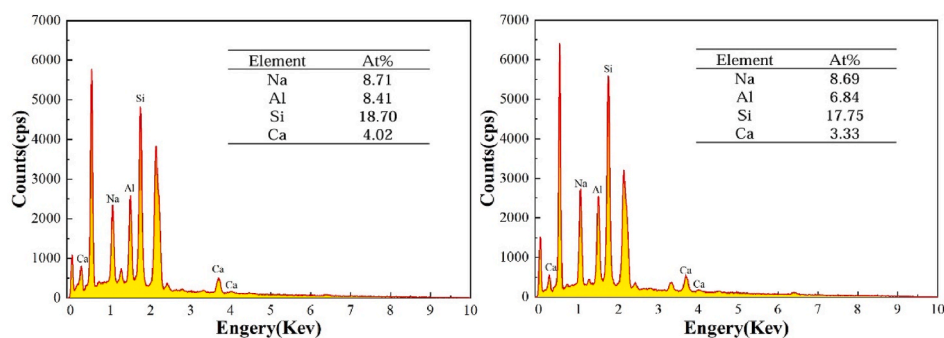
Fig. 14. SLSG and NS-SLSG magnified 2000 $\times$ and 500 $\times$ SEM images.



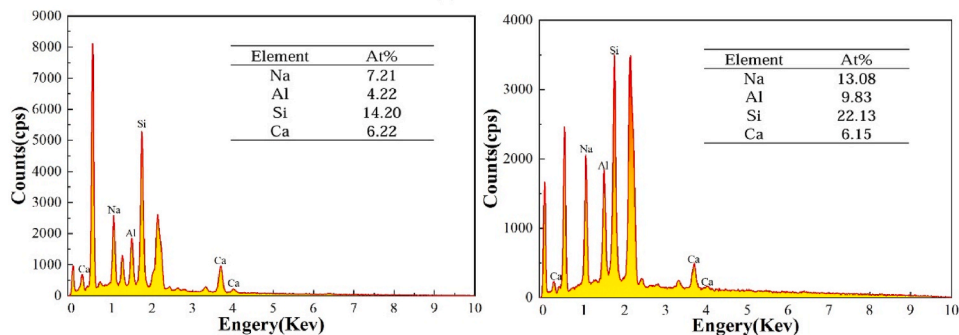
(a) P0



(b) P0.50



(c) P0.75



(d) P1.25

Fig. 15. EDS spectra of different regions of SLGS and NS-SLGS.

Table 6

Atomic ratios and standard deviations of EDS spectra in different regions of SLSG and NS-SLSG.

Sample	Al/Si	Na/Al	Ca/Si	Standard deviation		
				Al/Si	Na/Al	Ca/Si
P0	0.60, 0.93	1.50, 0.74	0.25, 0.01	0.17	0.38	0.12
P0.50	0.47, 0.56	0.33, 0.39	0.48, 0.25	0.05	0.03	0.12
P0.75	0.45, 0.38	1.04, 1.27	0.21, 0.19	0.04	0.12	0.01
P1.25	0.30, 0.44	1.71, 1.33	0.44, 0.28	0.07	0.19	0.08

inhomogeneous geopolymerization, resulting in a fragmented gel network. This lack of uniformity was associated with extensive cracking and a prevalence of weak gel layers, as shown in SEM images (Fig. 14(a)). Introducing an optimal NS dosage (0.75 wt%) enhances the uniformity of the geopolymerization process and promotes the formation of a continuous, dense gel network by optimizing elemental distribution, thereby improving mechanical properties. Conversely, the P1.25 group exhibits diminished enhancement efficacy due to NS agglomeration, with the Al/Si standard deviation (SD) increasing to 0.07. This suggests that excessive NS causes incomplete local reactions and the formation of a semi-dense structure, demonstrating the threshold effect of nanoparticle dispersion on reaction uniformity.

4. Conclusions

In this study, by systematically optimizing the preparation parameters of NS-SLSG and determining the optimal NS doping level, the flow properties, water recovery rate, mechanical properties, and mass strength efficiency of NS-SLSG were investigated. The enhancement mechanism of NS was elucidated through qualitative and quantitative microanalyses, culminating in the creation of a novel composite material suitable for in situ construction on the Moon. The major conclusions are as follows.

- (1) The best way to prepare NS-SLSG is to incorporate NS and alkali activators at the same time, and the water-binder ratio is 0.27. The inclusion of NS decreases the flowability of SLSG; however, acceptable flowability levels are maintained at moderate NS incorporation levels. Incorporating NS significantly reduces the water recovery rate of SLSG after 1 d, yet achieves over 96 % water recovery and enhances mechanical properties at 3 d.
- (2) A suitable amount of NS considerably enhances the mechanical properties of SLSG. The optimal NS doping level in NS-SLSG is 0.75 wt%, resulting in compressive and flexural strengths of 40.41 and 10.85 MPa, respectively. These strengths are 40.80 % and 57.02 % higher than those of the reference group, meeting the mechanical requirements for the primary load-bearing structures of the lunar research station.
- (3) NS incorporation endows the NS-SLSG group with greater mass-strength efficiency, significantly lowering the economic costs needed to achieve unit strength for lunar surface construction and reducing reliance on Earth's raw materials.
- (4) An appropriate proportion of NS notably decreases the total porosity (by 4.52 %) and pore volume of SLSG, increases bulk density, minimizes the number of coarse pores and microcracks, seals through cracks, and enhances pore structure. Simultaneously, NS can optimise the element distribution and promote the homogenization of the geopolymerization reaction. Furthermore, the nucleation effect induces the gel network to retain more chemically bound water, generate more product gels, and improve the thermal stability and gel structure.
- (5) NS functions as an active nanomaterial capable of inducing chemical bonding effects. It employs its internal Si-O-Si bonds as a structural backbone and, through surface hydroxyl groups, undergoes condensation reactions with depolymerized Al monomers. This process forms a highly polymerized gel structure, effectively balancing depolymerization and polycondensation rates to optimise the initial geopolymerization reaction.

CRediT authorship contribution statement

Zhenyu Wang: Writing – review & editing, Writing – original draft, Visualization, Formal analysis, Data curation. **Liqing Li:** Supervision, Methodology, Funding acquisition, Conceptualization. **Faping Li:** Validation, Supervision, Funding acquisition. **Banglu Xi:** Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

The authors gratefully acknowledge the support from the Henan Provincial Science and Technology Research Project (No. 222102320382), the Henan Provincial University Doctoral Program Fund (No. 2021BS034), the National Natural Science Foundation of China funded by the National Natural Science Foundation of China (No. 5240082149), and the National Natural Science Foundation

of China (Grant No. 52209127).

Data availability

Data will be made available on request.

References

- [1] G.G. Hady, C.D. Abigail, H. Sebastian, et al., ALCIDES: a novel lunar mission concept study for the demonstration of enabling technologies in deep-space exploration and human-robots interaction, *Acta Astronaut.* 151 (2018) 270–283.
- [2] China National Space Administration, International Lunar Research Station Partner Guide [EB/OL], 2021-06-16.
- [3] M. Isachenkov, S. Chugunov, I Akhatov, et al., Regolith-based additivemanu facturing forsustainable development of lunar infrastructure-Anoverview, *Acta Astronaut.* 180 (1) (2021) 650–678.
- [4] P. Zhang, W. Dai, R. Niu, et al., Overview of the lunar in situ resource utilization techniques for future lunar missions, *Space: Sci. Technol.* 3 (2023) 37.
- [5] G.X. Zhou, Z.Y. Cui, W.W. Zhang, et al., Conception of integrated scheme of “Exploration, mining and utilization of lunar water-ice resources, *Journal of Deep Space Exploration* 10 (2) (2023) 178–189.
- [6] F. Dang, Z. Wang, Y. Li, et al., Investigation of the melting behavior of laser-melted lunar regolith simulant for in-situ construction, *J. Mater. Res. Technol.* 36 (2025) 4574–4584.
- [7] Z. Wang, Z. Zheng, L. Jiang, et al., Microwave sintering of HUST-2 lunar highland regolith simulant: effects on microstructure evolution, mechanical and thermal properties, *Constr. Build. Mater.* 467 (2025) 140391.
- [8] H.A. Toutanji, S. Evans, R.N. Grugel, Performance of lunar sulfur concrete in lunar environments, *Constr. Build. Mater.* 29 (2012) 444–448.
- [9] J. Davidovits, *Geopolymer Chemistry and Reactions* [M], National Defense Industry Press, 2011.
- [10] W. Yunhua, L. Shiyong, Y. Pan, et al., Impact-related chemical modifications of the Chang'E-5 lunar regolith, *Geochem. Cosmochim. Acta* (2023), 36394–113.
- [11] S. Lee, A. Van Riessen, A review on geopolymer technology for lunar base construction, *Materials* 15 (13) (2022) 4516.
- [12] B. Lee, G. Kim, R. Kim, et al., Strength development properties of geopolymer paste and mortar with respect to amorphous Si/Al ratio of fly ash, *Constr. Build. Mater.* 151 (2017) 512–519.
- [13] G. Xiong, X. Guo, S. Yuan, et al., The mechanical and structural properties of lunar regolith simulant based geopolymer under extreme temperature environment on the moon through experimental and simulation methods, *Constr. Build. Mater.* 325 (2022) 126679.
- [14] K. Wang, P.N. Lemougna, Q. Tang, et al., Lunar regolith can allow the synthesis of cement materials with near-zero water consumption, *Gondwana Res.* 44 (2017) 1–6.
- [15] Z. Geng, Z. Wu, X. Wang, et al., A novel 3D printing scheme for lunar construction with extremely low binder utilization, *Addit. Manuf.* 99 (2025) 104657.
- [16] G. Davis, C. Montes, S. Eklund, Preparation of lunar regolith based geopolymer cement under heat and vacuum, *Adv. Space Res.* 59 (7) (2017) 1872–1885.
- [17] A. Alexiadis, F. Alberini, M.E. Meyer, Geopolymers from lunar and Martian soil simulants, *Adv. Space Res.* 59 (1) (2017) 490–495.
- [18] J.N. Mills, M. Katzarova, N.J. Wagner, Comparison of lunar and Martian regolith simulant-based geopolymer cements formed by alkali-activation for in-situ resource utilization, *Adv. Space Res.* 69 (1) (2022) 761–777.
- [19] S. Debbarma, X. Shi, A. Torres, et al., Fiber-reinforced lunar geopolymers synthesized using lunar regolith simulants, *Acta Astronaut.* 214 (2024) 593–608.
- [20] R. Zhang, S. Zhou, F. Li, et al., Mechanical and microstructural characterization of carbon nanofiber–reinforced geopolymer nanocomposite based on lunar regolith simulant, *J. Mater. Civ. Eng.* 34 (1) (2022) 04021387.
- [21] J.A. Abdalla, R.A. Hawileh, A. Bahurudeen, et al., Influence of synthesized nanomaterials in the strength and durability of cementitious composites, *Case Stud. Constr. Mater.* 18 (2023) e02197.
- [22] Z. Zidi, M. Ltifi, I. Zafar, Synthesis and attributes of nano-SiO₂ local metakaolin based-geopolymer, *J. Build. Eng.* 33 (2021) 101586.
- [23] P.S. Deb, P.K. Sarker, S. Barbhuiya, Sorptivity and acid resistance of ambient-cured geopolymer mortars containing nano-silica, *Cement Concr. Compos.* 72 (2016) 235–245.
- [24] F. Xu, J. Peng, R. Si, et al., Effect of synthesized sol-type Nano-SiO₂ on the mechanical properties and drying shrinkage of fly Ash-slag-based geopolymer, *J. Mater. Civ. Eng.* 36 (5) (2024) 04024082.
- [25] Z. Xu, H. Long, Q. Liu, et al., Mechanical properties and durability of geopolymer concrete based on fly ash and coal gangue under different dosage and particle size of nano silica, *Constr. Build. Mater.* 387 (2023) 131622.
- [26] T. Revathi, R. Jeyalakshmi, N.P. Rajamane, Study on the role of n-SiO₂ incorporation in thermo-mechanical and microstructural properties of ambient cured FA-GGBS geopolymer matrix, *Appl. Surf. Sci.* 449 (2018) 322–331.
- [27] X. Liu, Z.C. Peng, C.H. Pan, et al., Mechanical properties and microscopic analysis of nano-silica modified fly ash geopolymer, *Materials Reports* 34 (22) (2020) 22078–22082.
- [28] Z. Luo, W. Li, Y. Gan, et al., Nanoindentation on micromechanical properties and microstructure of geopolymer with nano-SiO₂ and nano-TiO₂, *Cement Concr. Compos.* 117 (2021) 103883.
- [29] K. Gao, K.L. Lin, D.Y. Wang, et al., Effects SiO₂/Na₂O molar ratio on mechanical properties and the microstructure of nano-SiO₂ metakaolin-based geopolymers, *Constr. Build. Mater.* 53 (2014) 503–510.
- [30] U. Durak, O. Karahan, B. Uzal, et al., Influence of nano SiO₂ and nano CaCO₃ particles on strength, workability, and microstructural properties of fly ash-based geopolymer, *Struct. Concr.* 22 (2021) E352–E367.
- [31] K. Sun, X. Peng, S. Wang, et al., Effect of nano-SiO₂ on the efflorescence of an alkali-activated metakaolin mortar, *Constr. Build. Mater.* 253 (2020) 118952.
- [32] A.M. Rashad, A.S. Ouda, Thermal resistance of alkali-activated metakaolin pastes containing nano-silica particles, *J. Therm. Anal. Calorim.* 136 (2019) 609–620.
- [33] H.M. Khater, Nano-silica effect on the physicomechanical properties of geopolymer composites, *Advances in nano research* 4 (3) (2016) 181.
- [34] H.U. Ahmed, A.A. Mohammed, A.S. Mohammed, Effectiveness of nano-SiO₂ on the mechanical, durability, and microstructural behavior of geopolymer concrete at different curing ages, *Arch. Civ. Mech. Eng.* 23 (2) (2023) 129.
- [35] M. Ibrahim, M.A.M. Johari, M. Maslehuddin, et al., Influence of nano-SiO₂ on the strength and microstructure of natural pozzolan based alkali activated concrete, *Constr. Build. Mater.* 173 (2018) 573–585.
- [36] M. Ibrahim, M. Nasir, S.R. Hussaini, et al., Performance of structurally viable green concrete derived from natural pozzolan and nanosilica, *Magazine of Civil Engineering* 107 (7) (2021), 10710–10710.
- [37] J. Shi, Y. Shen, W. Zhang, et al., Effects of three different nanomaterials on the properties and microstructure of sludge based geopolymers, *Constr. Build. Mater.* 414 (2024) 134965.
- [38] M. Jiang, L. Li, Y. Sun, Properties of TJ-1 lunar soil simulant, *J. Aero. Eng.* 25 (3) (2012) 463–469.
- [39] K. Zong, Z. Wang, J. Li, et al., Bulk compositions of the Chang'E-5 lunar soil: insights into chemical homogeneity, exotic addition, and origin of landing site basalts, *Geochem. Cosmochim. Acta* 335 (2022) 284–296.
- [40] C.R. Liu, S.J. Wang, J.M. Feng, et al., The geochemistry evidence for selecting the suitable raw material in China to simulate the low-titanium lunar soil from luna-sea basalt, *Mineral. Petrol.* (3) (2007) 28–33.
- [41] Z. Wu, Z. Geng, H. Pan, et al., Determinants of lunar geopolymer synthesis based on experimental and statistical analysis: towards lunar regolith's diversity, *Constr. Build. Mater.* 450 (2024) 138557.

- [42] J.P. Williams, D.A. Paige, B.T. Greenhagen, et al., The global surface temperatures of the moon as measured by the diviner lunar radiometer experiment, *Icarus* 283 (2017) 300–325.
- [43] R. Zhang, S. Zhou, F. Li, Preparation of geopolymer based on lunar regolith simulant at in-situ lunar temperature and its durability under lunar high and cryogenic temperature, *Constr. Build. Mater.* 318 (2022) 126033.
- [44] L. Li, Z. Shi, L. Wang, et al., Experimental study on rheological properties and 3D printing of simulated lunar soil polymers, *J. Build. Eng.* (2025) 112256.
- [45] GB/T 2419-2005, Test Method for Fluidity of Cement Mortar[S], China Standards Press, Beijing, 2005.
- [46] GB/T 17671-2021, Method of Testing cements-Determination of Strength (ISO Method)[S], China Standards Press, Beijing, 2021.
- [47] B. Ma, Z. Zhu, W. Huo, et al., Assessing the viability of a high performance one-part geopolymer made from fly ash and GGBS at ambient temperature, *J. Build. Eng.* 75 (2023) 106978.
- [48] Z. Geng, L. Zhang, H. Pan, et al., In-situ solidification of alkali-activated lunar regolith: insights into the chemical and physical origins, *J. Clean. Prod.* 391 (2023) 136147.
- [49] Z. Geng, L. Zhang, Z. Wu, et al., Silicate-lunar regolith composite: a vacuum self-hardened and comprehensively durable extraterrestrial construction material, *Constr. Build. Mater.* 449 (2024) 138517.
- [50] P. Zhang, Z. Gao, J. Wang, et al., Influencing factors analysis and optimized prediction model for rheology and flowability of nano-SiO₂ and PVA fiber reinforced alkali-activated composites, *J. Clean. Prod.* 366 (2022) 132988.
- [51] M. Liu, W. Dai, M. Li, et al., Mechanism of interface performance enhancement of nano-SiO₂ modified polyvinyl alcohol fiber reinforced geopolymer concrete: experiments, microscopic characterization, and molecular simulation, *J. Build. Eng.* 98 (2024) 111062.
- [52] S.M. Mustakim, S.K. Das, J. Mishra, et al., Improvement in fresh, mechanical and microstructural properties of fly ash-blast furnace slag based geopolymer concrete by addition of nano and micro silica, *Silicon* 13 (2021) 2415–2428.
- [53] W. Huo, Z. Zhu, H. Sun, et al., Reaction kinetics, mechanical properties, and microstructure of nano-modified recycled concrete fine powder/slag based geopolymers, *J. Clean. Prod.* 372 (2022) 133715.
- [54] X.Y. Zhu, Study on Early Reaction Mechanism of Metakaolin-based Geopolymer[D], Zhejiang University, Hangzhou, 2022.
- [55] C. Gunasekara, S. Setunge, D.W. Law, Long-term mechanical properties of different fly ash geopolymers, *ACI Struct. J.* 114 (3) (2017) 743.
- [56] G.L. Golewski, Investigating the effect of using three pozzolans (including the nanoadditive) in combination on the formation and development of cracks in concretes using non-contact measurement method, *Advances in Nano Research* 16 (3) (2024) 217–229.
- [57] G.L. Golewski, Using digital image correlation to evaluate fracture toughness and crack propagation in the mode I testing of concretes involving fly ash and synthetic nano-SiO₂, *Mater. Res. Express* 11 (9) (2024) 095504.
- [58] S. Zhou, C. Lu, X. Zhu, et al., Preparation and characterization of high-strength geopolymer based on BH-1 lunar soil simulant with low alkali content, *Engineering* 7 (11) (2021) 1631–1645.
- [59] G. Xue, G. Qiao, Impacts of thermal activation on lunar regolith simulant-based precursor and resulting geopolymer: composition, structure, solubility, and reactivity, *Cement Concr. Compos.* 155 (2025) 105840.
- [60] S. Zhou, X. Zhu, C. Lu, et al., Synthesis and characterization of geopolymer from lunar regolith simulant based on natural volcanic scoria, *Chin. J. Aeronaut.* 35 (1) (2022) 144–159.
- [61] F. Xu, G. Chen, K. Li, et al., Interfacial bond behavior between normal OPC concrete and self-compacting geopolymer concrete enhanced by nano-SiO₂, *Constr. Build. Mater.* 411 (2024) 134617.
- [62] L. Yang, Z. Zhu, D. Zhang, et al., Influence mechanism of Nano-SiO₂ on geopolymer recycled concrete: change mechanism of the microstructure and the anti-carbonation mechanism, *Cement Concr. Compos.* 146 (2024) 105364.
- [63] S. Guo, J. Tang, C. Xi, et al., Effect of retarders on setting and strength development of geopolymer activated by Na₂CO₃+Ca(OH)₂: properties and mechanisms, *Constr. Build. Mater.* 467 (2025) 140382.
- [64] F. Li, Q. Chen, Y. Lu, et al., Dry shrinkage and micro-structure of alkali-activated fly ash/slag pastes incorporated with silane coupling agent modified MWCNTs, *Cement Concr. Res.* 175 (2024) 107382.
- [65] G.U. Ryu, F. Pederson, Y. Oinam, et al., Characteristics of lunar geopolymer based on KLS-1 simulant considering particle size distribution, *ACS Earth Space Chem.* (2025).
- [66] I. Ismail, S.A. Bernal, J.L. Provis, et al., Modification of phase evolution in alkali-activated blast furnace slag by the incorporation of fly ash, *Cement Concr. Compos.* 45 (2014) 125–135.
- [67] Z. Yunsheng, S. Wei, C. Qianli, et al., Synthesis and heavy metal immobilization behaviors of slag based geopolymer, *J. Hazard Mater.* 143 (1–2) (2007) 206–213.
- [68] M. Rowles, B. O'connor, Chemical optimization of the compressive strength of aluminosilicate geopolymers synthesized by sodium silicate activation of metakaolinite, *J. Mater. Chem.* 13 (5) (2003) 1161–1165.
- [69] R. Sun, C. Fang, H. Zhang, et al., Chemo-mechanical properties of alkali-activated slag/fly ash paste incorporating white mud, *Constr. Build. Mater.* 291 (2021) 123312.
- [70] C.Y. Lai, A. Groth, S. Gray, et al., Preparation and characterization of poly (vinylidene fluoride)/nanoclay nanocomposite flat sheet membranes for abrasion resistance, *Water Res.* 57 (2014) 56–66.
- [71] B. Ren, J. Wang, Z. Zhou, et al., Regulation of the composition of metakaolin-based geopolymer: effect of zeolite crystal seeds, *Case Stud. Constr. Mater.* 19 (2023) e02421.
- [72] G. Xue, G. Qiao, Impacts of thermal activation on lunar regolith simulant-based precursor and resulting geopolymer: composition, structure, solubility, and reactivity, *Cement Concr. Compos.* 155 (2025) 105840.
- [73] X. Xi, Y. Zheng, J. Zhuo, et al., Influence of water glass modulus and alkali content on the properties of alkali-activated thermally activated recycled cement, *Constr. Build. Mater.* 452 (2024) 138867.
- [74] K.M. Klima, K. Schollbach, H.J.H. Brouwers, et al., Enhancing the thermal performance of class F fly ash-based geopolymer by sodalite, *Constr. Build. Mater.* 314 (2022) 125574.
- [75] X. Gao, Q.L. Yu, A. Lazaro, et al., Investigation on a green olivine nano-silica source based activator in alkali activated slag-fly ash blends: reaction kinetics, gel structure and carbon footprint, *Cement Concr. Res.* 100 (2017) 129–139.
- [76] Y. Luo, K.M. Klima, H.J.H. Brouwers, et al., Effects of ladle slag on class F fly ash geopolymer: reaction mechanism and high temperature behavior, *Cement Concr. Compos.* 129 (2022) 104468.
- [77] H. Lin, H. Liu, Y. Li, et al., Thermal stability, pore structure and moisture adsorption property of phosphate acid-activated metakaolin geopolymer, *Mater. Lett.* 301 (2021) 130226.
- [78] W. She, Z. Wu, J. Yang, et al., Cement-based biomimetic metamaterials, *J. Build. Eng.* 94 (2024) 110050.
- [79] X. Xi, Y. Zheng, J. Zhuo, et al., Mechanical properties and hydration mechanism of nano-silica modified alkali-activated thermally activated recycled cement, *J. Build. Eng.* 98 (2024) 110998.
- [80] G.L. Golewski, X. Xi, Y. Zheng, et al., Experimental evaluation of mode II fracture and microstructure of matrix-aggregate bond of concrete with crushed limestone, *Case Stud. Constr. Mater.* (2025) e05183.
- [81] G.L. Golewski, Effect of coarse aggregate type on the fracture toughness of ordinary concrete, *Infrastructure* 9 (10) (2024) 185.
- [82] H.F.W. Taylor, *Cement Chemistry*[M], Thomas Telford, London, 1997.